

8c

We can therefore conclude that our given the most conservative level of confidence at 99%, the true mean differs is still for $\log_{10}$ the 280

The conclusion is the same as in part a) - i.e. the two methods (C2 on one variable) get the same conclusion.

$$8b. P(0.86 \leq 226.22)$$

$$= P(226.22) - P(22 \leq 0.86) = 1 - 0.805 = 0.195$$

from normal

so $\beta = 0.195$ is our chance of a type 2 error

9a.

$$H_0: \mu = 280$$

$$n = 100$$

$$H_a: \mu > 280$$

$$\bar{x} = 358$$

$$s = 54$$

One sample z-test.

$$z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{358 - 280}{\frac{54}{\sqrt{100}}} = \frac{78}{5.4} = 14.44$$

Our critical value is $Z^{-1}(0.99) = 2.326$

Since our test statistic 14.44 is more than 2.326 the critical value we will reject H_0 and conclude there is sufficient evidence that that the mean calories is more than 280 calories

b. For a 1% CI:

$$\bar{x} - \text{margin of error} = 20.99 - SE$$

$$\text{The question asks for lower only so we take } 358 - 2.326(54/\sqrt{100}) = 345.44$$

so our 99% lower CI is $(345.44, \infty)$

c. The lower bound is 345.44 calories which is still far larger than the claim of 280 calories.